

# Dev Patel

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## EDUCATION

**M.S. Data Science**, California State University, East Bay — GPA 3.8/4.0

Aug 2024 – May 2026

**B.S. Mathematics**, California State University, Stanislaus — GPA 3.6/4.0

2019 – 2024

## EXPERIENCE

**Financial Analyst Intern**, City of Modesto, Modesto, CA

May 2024 – Aug 2024

- Analyzed budget reports and financial statements across 10+ city departments using SQL and Excel, surfacing discrepancies that informed annual budget reviews.
- Automated record retention and destruction workflows in Python, reducing unnecessary storage expenses by 20%.
- Built financial forecasting models in Python to project departmental expenditures, supporting budget-allocation decisions for multi-million-dollar city projects.

**Admissions Data Analyst**, California State University, Stanislaus, Turlock, CA

Dec 2023 – May 2024

- Analyzed close to 1,500 annual applications from students across 60+ countries using SQL and Excel to identify high-yield recruitment segments.
- Led data-driven marketing campaigns using predictive analytics and personalized outreach, driving a 10% increase in Spring 2024 enrollment and surpassing targets.
- Built interactive Tableau dashboards that delivered recruitment insights and secured a larger budget for international student marketing.

## PROJECTS

**Breast Cancer Diagnosis Using Supervised & Unsupervised Learning**

Aug 2024 – Dec 2024

- Built an L1-penalized SVM with 5-fold cross-validation for tumor classification, achieving 94% precision, 92% recall, and a 93% F1 score.
- Applied spectral clustering and K-means with majority-voting label assignment to reach an 88% F1 score (87% precision, 85% recall), validating stability via Monte Carlo simulation.

**Technologies:** Python, scikit-learn, TensorFlow, pandas, Matplotlib, Jupyter

**Agentic AI for Inventory Optimization** [\[GitHub\]](#)

Jan 2026 – May 2026

- Trained a LightGBM global forecasting model on 50M+ rows of Walmart M5 retail data, achieving a holdout RMSE of 1.79 ( $R^2$  0.65) and driving inventory policies that reduced simulated stockouts by 68% and improved ROI by 156%.
- Engineered a 500-path Monte Carlo simulation and a Mixed-Integer Linear Programming optimizer (Pyomo/HiGHS) to set cost-optimal (s, Q) policies under budget and service-level constraints, cutting portfolio operating cost by 25%.
- Integrated a context-grounded Claude Sonnet AI assistant into a multi-page Streamlit app, enabling managers to query results and run real-time demand-shock simulations in plain English.

**Technologies:** Python, pandas, scikit-learn, LightGBM, Pyomo, SciPy, NumPy, Streamlit, Plotly, Anthropic Claude API

**Traffic Forecasting for the City of San Francisco**

Dec 2024 – Feb 2025

- Built a traffic forecasting system using time-series models (ARIMA, ETS, SNaive, Prophet), achieving 90% accuracy over a 24-hour forecast horizon.
- Trained and validated models in R and Python with the fable and prophet packages, achieving an RMSE of 4.9 and an 18% reduction in MAPE.
- Processed 66,000+ traffic records from SF city roads, optimizing preprocessing and visualization for real-time traffic management.

**Technologies:** R, Python, fable, prophet, ARIMA, ETS, pandas, Seaborn

## TECHNICAL SKILLS

**Languages:** Python, SQL, R, Java, JavaScript

**Data Visualization & BI:** Tableau, Power BI, Excel, Plotly, Matplotlib, Seaborn

**ML & Data:** pandas, NumPy, scikit-learn, TensorFlow, PyTorch, LightGBM

**Databases & Cloud:** Microsoft SQL Server, MySQL, PostgreSQL, Azure Data Factory

**Coursework:** Quantitative Statistics, Optimization, Time-Series Analysis, Big Data Technologies, Database Management